

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS**  
**COURSE CODE: CSA0714**

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### **COURSE OUTCOME COVERED**

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

*Assessment Tool Weightage: 30%*

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QUESTIONS: 5

Annexure A

### **Scenario-Based**

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#### ***Code of Conduct:***

*I **Devi Shree Babu** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

### **Scenario-Based**

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based

on the expected number of devices, the following subnet masks have already been assigned:

School of Computing  $\rightarrow /25$

School of Electronics  $\rightarrow /26$

School of Mechanical Engineering  $\rightarrow /27$

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Answer:

### Subnet Allocation for Smart Campus Network

Given IP Address Block: 192.168.100.0/24

1. School of Computing – /25

Calculation:

$$\text{Number of host bits} = 32 - 25 = 7$$

$$\text{Total IP addresses} = 2^7 = 128$$

$$\text{Usable host addresses} = 128 - 2 = 126$$

Subnet range: 192.168.100.0 – 192.168.100.127

Network Address: 192.168.100.0

Broadcast Address: 192.168.100.127

Valid Host IP Address Range: 192.168.100.1 – 192.168.100.126

Number of Usable Host Addresses: 126

## 2. School of Electronics – /26

Calculation:

Number of host bits =  $32 - 26 = 6$

Total IP addresses =  $2^6 = 64$

Usable host addresses =  $64 - 2 = 62$

Subnet range: 192.168.100.128 – 192.168.100.191

Network Address: 192.168.100.128

Broadcast Address: 192.168.100.191

Valid Host IP Address Range: 192.168.100.129 – 192.168.100.190

Number of Usable Host Addresses: 62

## 3. School of Mechanical Engineering – /27

Calculation:

Number of host bits =  $32 - 27 = 5$

Total IP addresses =  $2^5 = 32$

Usable host addresses =  $32 - 2 = 30$

Subnet range: 192.168.100.192 – 192.168.100.223

Network Address: 192.168.100.192

Broadcast Address: 192.168.100.223

Valid Host IP Address Range: 192.168.100.193 – 192.168.100.222

Number of Usable Host Addresses: 30

#### 4. Remaining Unused IP Address Range

The allocated subnets use the IP addresses from 192.168.100.0 to 192.168.100.223.

The original network block ends at 192.168.100.255.

Therefore, the remaining unused IP address range is:

192.168.100.224 – 192.168.100.255

Total remaining IP addresses =  $255 - 224 + 1 = 32$  IP addresses.

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2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

Answer:

Subnet Allocation for Multinational Software Company

Given IP Address Block: 10.10.0.0/16

1. Software Development – /24

Calculation:

Number of host bits =  $32 - 24 = 8$

Total IP addresses =  $2^8 = 256$

Usable host addresses =  $256 - 2 = 254$

Subnet Range: 10.10.0.0 – 10.10.0.255

Network Address: 10.10.0.0

Broadcast Address: 10.10.0.255

Valid Host IP Address Range: 10.10.0.1 – 10.10.0.254

Number of Usable Host Addresses: 254

2. Quality Assurance – /25

Calculation:

Number of host bits =  $32 - 25 = 7$

Total IP addresses =  $2^7 = 128$

Usable host addresses =  $128 - 2 = 126$

Subnet Range: 10.10.1.0 – 10.10.1.127

Network Address: 10.10.1.0

Broadcast Address: 10.10.1.127

Valid Host IP Address Range: 10.10.1.1 – 10.10.1.126

Number of Usable Host Addresses: 126

### 3. Human Resources – /26

Calculation:

Number of host bits =  $32 - 26 = 6$

Total IP addresses =  $2^6 = 64$

Usable host addresses =  $64 - 2 = 62$

Subnet Range: 10.10.1.128 – 10.10.1.191

Network Address: 10.10.1.128

Broadcast Address: 10.10.1.191

Valid Host IP Address Range: 10.10.1.129 – 10.10.1.190

Number of Usable Host Addresses: 62

### 4. Executive Management – /27

Calculation:

Number of host bits =  $32 - 27 = 5$

Total IP addresses =  $2^5 = 32$

Usable host addresses =  $32 - 2 = 30$

Subnet Range: 10.10.1.192 – 10.10.1.223

Network Address: 10.10.1.192

Broadcast Address: 10.10.1.223

Valid Host IP Address Range: 10.10.1.193 – 10.10.1.222

Number of Usable Host Addresses: 30

#### 5. Unused IP Address Range

The allocated subnets use IP addresses from 10.10.0.0 to 10.10.1.223.

The original /16 network block ranges from 10.10.0.0 to 10.10.255.255.

Therefore, the remaining unused IP address range is:

10.10.1.224 – 10.10.255.255

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3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

**2001:0db8:abcd:0000:0000:0000:0000/64**

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.

2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

Answer:

### IPv6 Addressing for Smart Hospital Network

Given IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Convert the Given IPv6 Address into Compressed Notation

Given address:

2001:0db8:abcd:0000:0000:0000:0000:0000

The consecutive groups of zeros can be replaced by ::.

Compressed IPv6 Address:

2001:db8:abcd::/64

2. Expand the Compressed Address into Full Hexadecimal Representation

Compressed Address:

2001:db8:abcd::/64

Expanded Full IPv6 Address:



2001:0db8:abcd:0000:0000:0000:0000/64

### 3. Identify the Network Prefix

Prefix Length = /64

Therefore, the first 64 bits represent the network portion.

Network Prefix:

2001:0db8:abcd:0000::/64

### 4. Calculate the Total Number of Host Addresses Supported

IPv6 address length = 128 bits

Network bits = 64 bits

Host bits =  $128 - 64 = 64$  bits

Number of host addresses =  $2^{64}$

Therefore, the subnet supports:

18,446,744,073,709,551,616 host addresses.

### 5. Total Number of Possible Host Addresses Available within the Network

The total number of possible addresses in a /64 IPv6 network is:

$2^{64} = 18,446,744,073,709,551,616$  addresses.

Unlike IPv4, IPv6 does not reserve separate network and broadcast addresses.

Therefore, the total number of possible addresses is:

18,446,744,073,709,551,616 addresses.

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4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

**192.168.2.45**

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

Answer:

Routing Table Analysis

Given Routing Table Entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP Address: 192.168.2.45

1. Destination Subnet

The destination IP address is 192.168.2.45.

The first three octets are 192.168.2, which matches the subnet:

192.168.2.0/24

Therefore, the packet belongs to the destination subnet:

192.168.2.0/24

## 2. Matching Routing Table Entry

The matching routing table entry is:

192.168.2.0/24

## 3. Next-Hop Network/Interface Used to Forward the Packet

The router forwards the packet through the interface connected to:

192.168.2.0/24

Therefore, the next-hop network/interface is:

192.168.2.0/24 interface

## 4. Default Route

If the destination IP address does not match any of the routing table entries, the router uses the default route:

0.0.0.0/0

This route forwards packets to the configured default gateway or next-hop router.

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5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Answer:

### IP Addressing for Two Office Departments

Given Subnet Information:

Administration LAN: 192.168.10.0/26

Finance LAN: 192.168.10.64/26

1. Administration LAN – 192.168.10.0/26

Calculation:

Number of host bits =  $32 - 26 = 6$

Total IP addresses =  $2^6 = 64$

Usable host addresses =  $64 - 2 = 62$

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Valid Host IP Address Range: 192.168.10.1 – 192.168.10.62

Suitable IP Addresses for Three Computers:

Computer 1: 192.168.10.2

Computer 2: 192.168.10.3

Computer 3: 192.168.10.4

Default Gateway: 192.168.10.1

2. Finance LAN – 192.168.10.64/26

Calculation:

Number of host bits =  $32 - 26 = 6$

Total IP addresses =  $2^6 = 64$

Usable host addresses =  $64 - 2 = 62$

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Valid Host IP Address Range: 192.168.10.65 – 192.168.10.126

Suitable IP Addresses for Three Computers:

Computer 1: 192.168.10.66

Computer 2: 192.168.10.67

Computer 3: 192.168.10.68

Default Gateway: 192.168.10.65

### 3. Default Gateway for Communication

Administration LAN hosts should use 192.168.10.1 as their default gateway.

Finance LAN hosts should use 192.168.10.65 as their default gateway.

The Layer-3 router uses these gateway addresses to route traffic between the Administration LAN and Finance LAN.

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## RUBRICS FOR CALCULATION

| Scenario Based Assessment    |               |                                                                           |                                                 |                                                   |                                                   |
|------------------------------|---------------|---------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Criteria                     | Weightage (%) | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                            | Level 2 (Developing)                              | Level 1 (Beginning)                               |
| Understanding of Scenario    | 20%           | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps        | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20%           | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |
| Application of Concepts      | 25%           | Applies appropriate concepts                                              | Applies concepts                                | Limited or partially                              | Unable to apply                                   |

|                     |     |                                                               |                                                    |                                                 |                                               |
|---------------------|-----|---------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------|-----------------------------------------------|
|                     |     | effectively to address the scenario                           | with minor errors                                  | correct application                             | relevant concepts                             |
| Decision Making     | 20% | Makes well-reasoned , logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented        | Generally clear with minor issues                  | Somewhat unclear or poorly structured           | Disorganized and difficult to understand      |